

Simplicial compactification and pseudodifferential
operators
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Abstract

I will start from the notion of a generalized product and the associated 'small' algebra of pseudodifferential operators and 'large' modules over it. After describing some of the properties and examples I will pass to the special case of the (in general conjectural) simplicial compactification of a Lie group. As time permits I will indicate possible implications of such a construction. This is based in part on joint work with Pierre Albin and with Chris Kottke and Michael Singer.

Generalized products

- ▶ The notion of a *generalized product* is intended as an organizing principle for the construction of algebras (and modules) of pseudodifferential operators on manifolds with corners, although I do not really know the limits of its applicability.
- ▶ It is convenient to express the definition in the language of categories, see [4].
- ▶ As source take the small category with objects, $\mathbb{N}$, the set of natural numbers, and morphisms from n to m consisting of all the maps $J(n) \longrightarrow J(m)$, $J(n) = \{1, 2, \dots, n\} \subset \mathbb{N}$.
- ▶ This is equivalent to, indeed is a ‘skeleton’ of, the category of finite sets and maps between them.

Manifolds with corners

- ▶ As target consider the category with objects, $\mathcal{M}$, the smooth manifolds, here meaning manifolds with corners, and as morphism the b-maps. The main interest lies in the case of *compact* manifolds.
- ▶ There are other options for the target, such as manifolds with 'dividers', which are internal normally intersecting submanifolds. Also in the complex case manifolds with divisors.
- ▶ Manifolds with corners are locally modeled on open sets in $[0, \infty)^d$ and each point maps to the origin in a coordinate neighbourhood in $[0, \infty)^l \times \mathbb{R}^{d-l}$.
- ▶ A smooth map in coordinates $f : [0, \infty)_x^l \times \mathbb{R}^{d-l} \supset \Omega \longrightarrow [0, \infty)_{x'}^{d'} \times \mathbb{R}^{d'-l'}$ is a b-map if

$$f^*(x'_j) = a_j x^\alpha, \quad 0 < a_j \in \mathcal{C}^\infty \text{ or } 0. \quad (1)$$

Here α is a multiindex, possibly zero. These maps are well-behaved with respect to the boundaries.

Generalized products defined

- ▶ A generalized product is a contravariant functor $F : \mathbb{N} \longrightarrow \mathcal{M}$ with three additional properties.
- ▶ The underlying map F gives a sequence of manifolds, which I write as $M[k]$.
- ▶ Then as a functor each map $\sigma : J(n) \longrightarrow J(m)$ has image a b-map $S_\sigma : M[m] \longrightarrow M[n]$ (with the composition properties required of a functor).
- ▶ The additional conditions are:
 1. The dimension is linear in k :

$$\dim M[k] = p + (k - 1)\alpha. \quad (2)$$

2. If σ is injective then S_σ is a b-fibration.
3. If σ is surjective then S_σ is a p-embedding.

b-fibrations and p-embeddings

A map $M \longrightarrow N$ of compact manifolds (with corners)

- ▶ Is a fibration if it is surjective and the pull-back of coordinates are locally a subset of coordinates.
- ▶ Is a b-fibration if this is *weakened* to the pull-backs being a subset of tangential coordinates and powers x^α of boundary defining functions where the different α s have no common non-zero term.
- ▶ Is a p-embedding if it is a diffeomorphism onto its image which must be locally defined by the vanishing of a subset of coordinates.
- ▶ A map is both a b-fibration and a p-embedding iff it is a diffeomorphism.

The image of a p-embedding has a normal *neighbourhood*.

Quantization

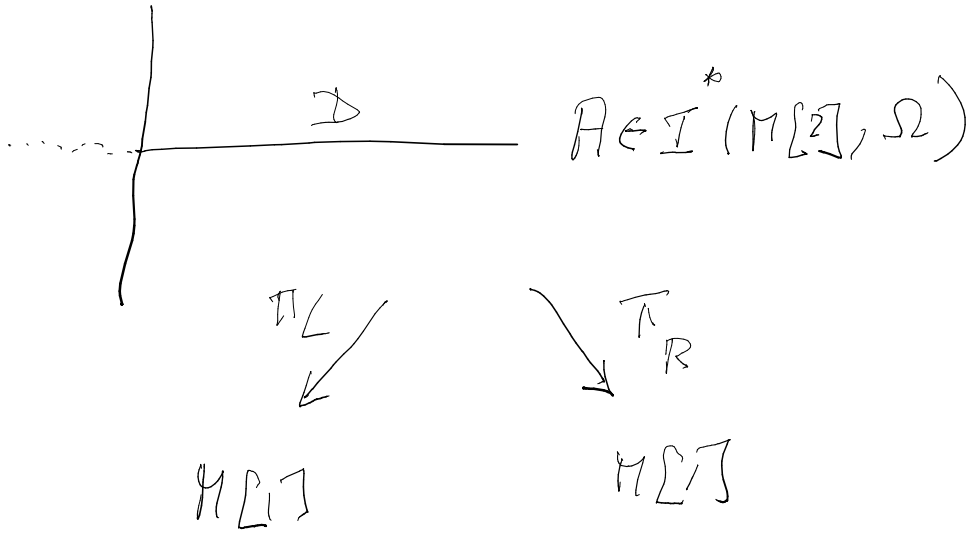
At least one of the reasons for introducing this notion is:

Theorem

A generalized product defines a Lie algebroid over $M[1]$ and a 'quantization' of it to an algebra of 'pseudodifferential operators' $\Psi^(M[*])$ and modules $\Psi^{*,*}(M[*])$.*

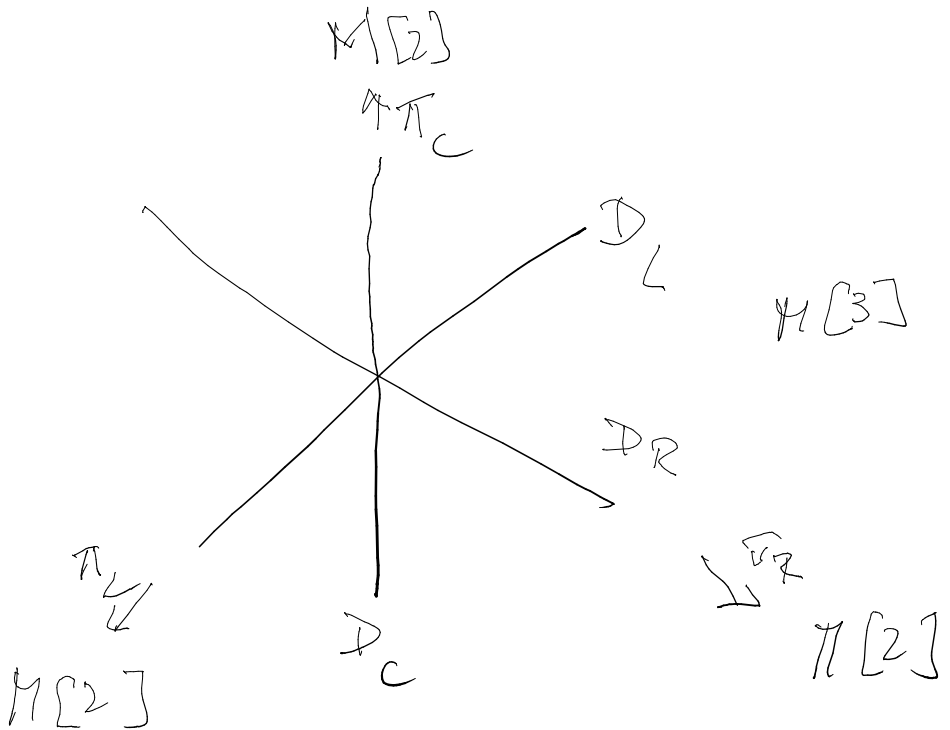
- ▶ The pseudodifferential operators act on $\mathcal{C}^\infty(M[1])$ but the action is not always faithful; the product gives a faithful action on $\Psi^{-\infty}(M[*]) \simeq \dot{\mathcal{C}}^\infty(M[2])$.
- ▶ Generalized products are related to, but significantly they are generally *not*, groupoids. Fundamentally this difference arises from the presence of boundaries and the assumption of compactness, both central here. A non-compact group can never have a compactification which is a group and, except in rather trivial cases, this carries over to groupoids.
- ▶ In view of this I do not think 'groupoid' is the appropriate concept in this context.

Definition of $\Psi^*(M[\mathbb{I}]) \ni A$



$$\bar{\Psi}^{\mu_1}(\pi[1]) \cdot \bar{\Psi}^{\mu_2}(\pi[1])$$

$$\subset \bar{\Psi}^{\mu_1 + \mu_2}(\pi[1])$$



$$A_1 \cdot A_2 = (\pi_C)_* (\pi_L^* A_1 \cdot \pi_R^* A_2)$$

Lie algebroid

- ▶ The Lie algebroid resolved by a generalized product is defined in essentially the same way as for a groupoid.
- ▶ Namely there are two structure maps $\Pi_L, \Pi_R : M[2] \longrightarrow M[1]$ associated to the two maps $\{1\} \rightarrow \{1, 2\}$ and by definition these are b-fibrations.
- ▶ There is a diagonal submanifold $S_D : M[1] \longrightarrow M[2]$ associated to the unique map $D : \{1, 2\} \longrightarrow \{1\}$. It follows that Π_L and Π_R are fibrations near the image $M[1] \equiv D \subset M[2]$ transversal to D .
- ▶ The 'structure Lie algebroid' is the vector bundle

$$V = \text{null}((\Pi_R)_*) \subset {}^b TM[2]. \quad (3)$$

- ▶ The fact that it is a Lie algebra is less obvious and depends on the properties of $M[3]$.
- ▶ The anchor map is $(\Pi_L)_* : V \longrightarrow {}^b TM[1] \longrightarrow TM[1]$ which maps sections of V to b-tangent vector fields on $M[1]$.
- ▶ The null space of the anchor map at each point is always a Lie algebra and is non-trivial at any boundary point of $M[1]$.

Products

- ▶ The core example is the powers, $M[k] = M^k$, of a (compact) manifold without boundary where the 'structure maps' S_σ just correspond to factor-rearrangement

$$S_\sigma(q_1, \dots, q_m) = (q_{\sigma(1)}, \dots, q_{\sigma(m)}). \quad (4)$$

- ▶ This is an embellishment of what is, in some quarters, called the 'pair groupoid'.
- ▶ The injective maps $J(n) \rightarrow J(m)$, $m > n$, correspond to projections $M^m \rightarrow M^n$ and the surjective maps $J(n) \rightarrow J(m)$, $m < n$, correspond to embeddings $M^m \rightarrow M^n$ as 'multidiagonals'.
- ▶ The proof of the theorem above basically reduces to understanding the geometry of a generalized product and applying a variant of the construction on manifolds without boundary.

Normal operator

- ▶ One of the fundamental properties of a generalized product is that to each boundary hypersurface H of $M[k]$ there is an associated generalized product $H[k]$ consisting of boundary hypersurfaces of $M[k]$.
- ▶ This generates a surjective homomorphism, the normal homomorphism at H , of the associated algebras of pseudodifferential operators

$$N_H : \Psi^m(M[*]) \longrightarrow \Psi^m(H[*]). \quad (5)$$

- ▶ One of the aims of my talk is to discuss examples of the structure of N_H and extensions which stem from this.

Shifting

- ▶ I'm not sure that many people buy into the utility of generalized products.
- ▶ In particular the question arises of the utility of the spaces $M[k]$ for $k > 3$?
- ▶ Here is one observation that actually has geometric applications.
- ▶ One can embed $J(k)$ in $J(k+1)$ by say the increasing embedding, so just $k+1$ is missing from the range.
- ▶ This gives a functor $\mathbb{N} \rightarrow \mathbb{N}$, mapping n to $n+1$ and embedding maps $J(n) \rightarrow J(m)$ in $J(n+1) \rightarrow J(m+1)$ by always mapping the last element to the last.
- ▶ It follows that $M[*+1]$ acquires the structure of a generalized product by forgetting data.
- ▶ In the algebra of pseudodifferential operators, the last variable is now a 'parameter' over the interior in the initial calculus.
- ▶ So the operators 'act on the fibres' but these are actually the fibres of a b-fibration, not a fibration.

Examples

- ▶ First let me briefly recall some examples of generalized products. I will limit these to the best-known examples where $M[1]$ is a (compact) manifold with boundary.
- ▶ There is an intrinsic invariance which means that if we have a fibre bundle $M \longrightarrow Y$ in the category of manifolds with corners, so locally trivial, and we are given a generalized product on one of the fibres, then by local trivialization we can extend it to a fibrewise structure which is also a generalized product. So I ignore this overall possibility of ‘families’.
- ▶ If M is a compact manifold with boundary then there is a generalized product (not necessarily uniquely) associated to a Lie algebra $\mathcal{V} \subset \mathcal{V}_b(M)$ provided it satisfies fibration/filtration conditions.
- ▶ It is required to satisfy the finiteness condition $\mathcal{V} \supset x^p \mathcal{V}_b$ for some p , where x is a boundary defining function.

Local structure

- ▶ The local structure of $\mathcal{V}$ required is that $\mathcal{V}$ have a basis

$$x^{r+1}\partial_x, \partial_{y_j}, xV_*^{(1)}, x^2V_*^{(2)}, \dots, x^N V_*^{(N)}. \quad (6)$$

- ▶ Any of the powers may be missing.
- ▶ The ∂_{y_j} should span the fibres of a fibration of the boundary, really corresponding to a subbundle of ${}^bT_{\partial M}M$.
- ▶ Each of the $V_*^{(l)}$ span a subbundle of ${}^bT_{\partial M}M$ which together form a Lie filtration.
- ▶ Given these conditions there is an associated generalized product which resolves $\mathcal{V}$ in the sense that it is the associated Lie algebroid and hence quantizes it.
- ▶ The naming conventions for these algebras, and generalized products, is complicated at best.
- ▶ I will ignore history and just say r is the *order* of the structure and N is the *degree*.

Semiclassical case

- ▶ The case in which the ‘normal’ term $x^{r+1}\partial_x$ is missing is however special since we cannot then have the condition $\mathcal{V} \supset x^p \mathcal{V}_b(M)$.
- ▶ Rather the requirement is that $M = N \times [0, 1]$ and $\mathcal{V} \supset x^p \mathcal{V}(N)$ the space of x -dependent vector fields on N . This is then an ‘iterated semiclassical/adiabatic’ calculus.

Higher codimension

- ▶ There are many examples for manifolds with corners. I hope that a full description of these will be forthcoming!
- ▶ The calculus (but not I think the full generalized product) has been discussed in the case of iterated conic spaces and many body spaces.
- ▶ I want to caution that the restrictions on $\mathcal{V}$ have not been fully explored in the case of corners. In particular there is the question of 'iteration'.
- ▶ Imagine a manifold with corners equipped with structures such as in (6) on the interior of each boundary hypersurface. Initially just consider the fibrations. Clearly smoothness up to corners plus some compatibility condition is needed. The usual requirement is iteration, that one fibration is finer than the other.

$SL(3, \mathbb{R})$

- ▶ An instructive (counter)example is the hd-compactification of $SL(3, \mathbb{R})$. This is a compact manifold with corners of codimension two with two boundary hypersurfaces which are diffeomorphic and each carry a fibration, so these are of the same rank.
- ▶ At the interiors of the hypersurfaces this is a boundary structure of order zero and weight one.
- ▶ They are compatible at the corner in the sense that the intersection of the two fibrations is a fibration but it is strictly smaller than either.
- ▶ In his thesis Malte Behr [1] constructed a calculus by first resolving the corner, so giving iterated fibrations, but this increases the weight on the new hypersurface to two.
- ▶ However, one can construct a generalized product with single space the unresolved compactification and the question remains as to whether this has 'good properties' the form of which I will mention below.

Semiclassical extension

- ▶ One universal property is that every generalized product has a semiclassical extension with single space $M[1] \times [0, 1]_\epsilon$.
- ▶ That is, if V is the Lie algebroid bundle of $M[k]$ then the structure bundle of the semiclassical extension $M[*, \text{sl}]$ is the rescaling of ϵV to a subbundle of the tangent bundle at the diagonal.
- ▶ It is obtained by blowing up all the submanifolds $\mathcal{D} \times \{0\}$ where $\mathcal{D}$ is the collection of multidiagonals in $M[k]$ for each k .

Order zero and weight one

- ▶ Consider the basic case where $r = 0$, $N = 1$ in (6), $[5, 2, 3]$, variously called complete edge (from the metric) or fibred boundary and determined by a fibration of the boundary $\phi : \partial M \longrightarrow B$.
- ▶ The associated Lie algebra of vector fields is

$$\mathcal{V}_{\phi b}(M) = \{V \in \mathcal{V}_b(M); V \text{ is tangent to the fibres of } \phi\}. \quad (7)$$

- ▶ This structure has a resolution to a generalized product defined by iterated blow-up

$$M[k, \phi b] = [M^k; \{\mathcal{C}_{\mathcal{P}}\}]. \quad (8)$$

- ▶ Here the centres of resolution $\mathcal{C}_{\mathcal{P}} \subset M^k$ are parameterized by partitions of $J(k)$ and are each the closure of the flow-outs with respect to $\mathcal{V}_{\phi b}$, in each factor, of the boundary of the multi-diagonal defined by $\mathcal{P}$. They should (and can) be blown up in order of increasing dimension.

Zero calculus

- ▶ To understand the normal homomorphism for the case, consider the extreme 'zero structure' where the fibration ϕ is the point fibration.
- ▶ Then $\mathcal{V}_{\phi b}(M) = \mathcal{V}_0(M)$ is the Lie algebra of smooth vector fields on M which vanish at the boundary, in the ordinary sense.
- ▶ In this case the centers of blow-up in (8) are simply the boundaries of all the multi-diagonals.
- ▶ The manifold $H[2]$, fibres over H with fibres which are quarter spheres. The interiors of these fibres are Lie groups, isomorphic to the sovable group

$$G = \mathbb{R}^+ \ltimes \mathbb{R}^{d-1}. \quad (9)$$

- ▶ Thus fibres of $H[2]$ are compactifications of G and together with the fibres of the higher $H[k]$ form a generalized product which we denote $\overline{G}[k, \text{SC}]$

$$\dim \overline{G}[k, \text{SC}] = (k-1)d. \quad (10)$$

Fibres of $H[k]$

- ▶ The manifolds in the generalized product (10) are compactifications of G^{k-1} .
- ▶ This can be understood more generally starting from a Lie group G and passing from the products G^k to the quotients by the diagonal right action of G

$$G^k/G \simeq G^{k-1}. \quad (11)$$

- ▶ The product structure on G^k descends to a generalized product structure on the quotient. So on G , as the double space, the reflection (corresponding to the non-trivial involution on $J(2)$) is inversion.
- ▶ Similarly on G^2 , as the triple space, the quotient action of Σ_3 is generated by the three involutions

$$(g_1, g_2) \longmapsto (g_2, g_1), (g_1^{-1}, g_2 g_1^{-1}), (g_1 g_2^{-1}, g_2^{-1}). \quad (12)$$

Simplicial compactifications

Definition

A simplicial compactification of a Lie group G is a generalized product, $\overline{G}[k, SC]$, given by compactifications (to manifolds with corners) of the quotient generalized product (11).

- ▶ The obvious question is, do they exist?
- ▶ The construction of the zero stretched product implies that they do exist for the solvable groups (9).
- ▶ Similar constructions hold for more general solvable groups which are semi-direct products of $\mathbb{R}^+$ with filtered nilpotent rather than abelian factors.
- ▶ Why is this relevant here? Recall for a generalized product there are k maps $\Pi_I : X[k] \longrightarrow X[1]$ and hence a smooth map $\Pi_{J(k)} : X[k] \longrightarrow X[1]^k$.

Suspension

Theorem

*If P is a smooth, locally trivial, bundle of Lie groups, modeled on G , over $X[1]$ for a generalized product and G has a simplicial compactification then the pull back $(\Pi_{J(k)})^*P$ of P has a natural compactification, with fibres $\overline{G}[k]$, to a 'suspended' generalized product*

$$\overline{G}[k] \# X[k]. \quad (13)$$

- For the order zero, weight one product $M[k, \phi]$ described above, the boundary generalized product is $\overline{G}[k] \# B[k, \phi]$ where $B[k, \phi]$ is the generalized product formed by the fibre products of $\phi : \partial M \longrightarrow B$ and $\overline{G}[k]$ is the simplicial compactification (10) of the solvable group (9).

Final conjecture

Conjecture

Every semisimple Lie group has a simplicial compactification.

- ▶ Pierre Albin and I have shown this for the real rank one groups $SO^+(1, n)$ and the argument probably extends to the other semisimple groups of real rank one.
- ▶ What would this buy you? A large convolution module which should be useful for instance in treating left-invariant Laplacians.
- ▶ The spectral decomposition for such an operator corresponds to the Plancherel decomposition of $L^2(G)$.
- ▶ One might expect that the representations of G which appear in $L^2(G)$ should be parameterized by kernels on $\overline{G}[2, SC]$ and $\overline{G}[3, SC]$.



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